Matrix Inversion using Gauss Jordan Elimination

$$\left[\begin{array}{ccc|ccc} -3 & 2 & -1 & 1 & 0 & 0 \\ 6 & -6 & 7 & 0 & 1 & 0 \\ 3 & -4 & 4 & 0 & 0 & 1 \end{array}\right] \quad \begin{array}{l} r_2 = r_2 + 2r_1 \\ r_3 = r_3 + r_1 \end{array}$$

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$$\left[\begin{array}{ccc|ccc} -3 & 2 & -1 & 1 & 0 & 0 \\ 0 & -2 & 5 & 2 & 1 & 0 \\ 0 & -2 & 3 & 1 & 0 & 1 \end{array}\right] \quad r_3 = r_3 - r_2$$

$$\left[\begin{array}{ccc|ccc} -3 & 2 & -1 & 1 & 0 & 0 \\ 0 & -2 & 5 & 2 & 1 & 0 \\ 0 & 0 & -2 & -1 & -1 & 1 \end{array}\right] \quad \begin{array}{l} r_1 = r_1 - \frac{1}{2} r_3 \\ r_2 = r_2 + \frac{5}{2} r_3 \end{array}$$

$$\left[\begin{array}{ccc|ccc} -3 & 2 & 0 & 3/2 & 1/2 & -1/2 \\ 0 & -2 & 0 & -1/2 & -3/2 & 5/2 \\ 0 & 0 & -2 & -1 & -1 & 1 \end{array}\right] \quad r_1 = r_1 + r_2$$

$$\left[\begin{array}{ccc|ccc} -3 & 0 & 0 & 1 & -1 & 2 \\ 0 & -2 & 0 & -1/2 & -3/2 & 5/2 \\ 0 & 0 & -2 & -1 & -1 & -1 \end{array}\right] \quad \begin{array}{l} r_1 = -r_1/3 \\ r_2 = -r_2/2 \\ r_3 = -r_3/2 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1/3 & 1/3 & -2/3 \\ 0 & 1 & 0 & 1/4 & 3/4 & -5/4 \\ 0 & 0 & 1 & 1/2 & 1/2 & -1/2 \end{array}\right]$$

$$\underbrace{\qquad\qquad}_{I} \qquad \underbrace{\qquad\qquad}_{A^{-1}}$$